

CHAPTER 2- STATISTICAL ESTIMATIONS

2.1. Basic Concepts

There are two types of inference: **estimation** and **hypothesis testing**; **Estimation** – is a process whereby we select a **random sample** from a population and use a **sample statistic** to estimate a **population parameter**. The objective of estimation is to determine the *approximate value* of a population parameter on the basis of a sample statistic.

E.g., the sample mean ($\bar{x}$) is employed to *estimate* the population mean (μ).

Inferential statistics is concerned with estimation. In many cases values for a population parameter are unknown. If parameters are unknown it is generally not sufficient to make some convenient assumption about their values, rather those unknown parameters should be estimated.

In business many decision are made without complete information.

A firm does not know exactly what will be its sales volume next year or next month. A college does not know exactly how many students will enroll next year. Both must estimate to make decision about the future.

2.1 Types of Estimates

2.1.1 Point estimate

A number or a simple number is used to estimate a population parameter.

A random sample of observations is taken from the population of interest and the observed values are used to obtain a point estimate of the relevant parameter.

a. The sample mean, $\bar{x}$, is the best estimator of the population mean μ .

Different samples from a population yield different point estimates of μ ,

b. Sample proportion $\bar{p}$ is a good estimator of population proportion, p .

- Population proportion P is equal to the number of elements in the population belonging to the

category of interest divided by the total number of elements in the population $p = \frac{X}{N}$

Where: X is the number of success in the population and

N population size

Sample proportion, $\bar{p} = \frac{x}{n}$ where;

x is the number of elements in the sample found to belong to the category of interest and n is the sample size.

or $\bar{p} = \frac{\text{Number of success in a sample}}{\text{number sampled}}$

Example: of 2000 persons sampled 1600 favored more strict environmental protection measures, what is the estimated population proportion.

$$\bar{p} = \frac{1600}{2000} = 0.80$$

80% is an estimate of the proportion in the population that favor more strict measures

In general:

The statistic $\bar{x}$ estimates μ

S estimates σ

S^2 estimates σ^2

$\bar{p}$ estimates p

Estimators and their properties / Goodness of an estimator/

The properties of good estimators are

- | | |
|-----------------|--------------------|
| a) Unbiasedness | c) Consistency and |
| b) Efficiency | d) Sufficiency |

a) An estimator is said to be unbiased if its expected value is equal to the population parameter it estimates.

$E(\bar{x}) = \mu$ The sample mean, $\bar{x}$, is therefore, an unbiased estimator of the population mean. Any systematic deviation of the estimator away from the parameter of interest is called Bias.

b) An estimator is efficient if it has a relatively small variance (as standard deviation). The sample means have a variance of σ^2/n value is less than σ^2 . So the sample mean is an efficient estimator of the population mean.

c) An estimator is said to be consistent if its probability of being close to the parameter it estimates increases as the sample size increases.

The sample mean is a consistent estimator of μ . This is so because the standard deviation of $\bar{x}$ is

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. As the sample size n increases, the standard deviation of $\bar{x}$ decreases and hence the probability that $\bar{x}$ will be close to its expected value, μ , increases.

d) An estimator is said to be sufficient if it contains all the information in the data about the parameter it estimates. The sample mean is sufficient estimator of μ . Other estimators like the median and mode do not consider all values. But the mean considers all values (added and divided by the sample size).

2.1.2 Interval Estimates

Interval estimate **states the range** within which a population parameter probably lies. The interval within which a population parameter is expected to lie is usually referred to as the **confidence interval**.

The confidence interval for the population mean is the interval that has a high probability of containing the population mean, μ .

Two confidence intervals are used extensively.

1. 95% confidence interval and
2. 99% confidence interval

A 95% confidence interval means that about 95% of the similarly constructed intervals will contain the parameter being estimated. If we use the 99% confidence interval we expect about 99% of the intervals to contain the **parameter being estimated**.

Another interpretation of the 95 % confidence interval is that 95 % of the sample means for a specified sample size will lie **within 1.96 standard** deviations of the **hypothesized population** mean. For 99% the sample means will lie, within **2.58** standard deviations of the hypothesized population mean.

Where do the values 1.96 and 2.58 come from?

The middle 95% of the sample mean lie **equally on either side of the mean**. And logically $0.95/2=0.4750$ or 47.5% of the area is to the right of the mean and the area to the left of the mean

is 0.4750.

The Z value for this probability is 1.96.

The Z to the right of the mean is + 1.96 and Z to the left is – 1.96.

2.2.1 Constructing Confidence Interval

a) Compute the standard error of the mean

Standard error of the mean is the standard deviation of the sample means.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

σ = population standard deviation
 n = sample size

If the population standard deviation is not known, the standard deviation of the samples, is used

to approximate the population standard deviation. $S_{\bar{x}} = \frac{S}{\sqrt{n}}$

This indicates that the **error in estimating the population mean** decreases as the **sample size increases**.

b) The 95% and 99% confidence intervals are constructed as follows when $n \geq 30$.

$$\text{95\% confidence interval } \bar{x} \pm 1.96 \frac{S}{\sqrt{n}}$$

$$\text{99\% confidence interval } \bar{x} \pm 2.58 \frac{S}{\sqrt{n}}$$

1.96 and 2.58 indicate the Z values corresponding to the middle 95% or 99% of the observation respectively.

In general a confidence interval for the mean is computed by $\bar{x} \pm Z \frac{S}{\sqrt{n}}$, Z reflects the selected level of confidence.

Example. An experiment involves selecting a random sample of 256 middle managers for studying their annual income. The sample mean is computed to be Br. 35,420 and the sample standard deviation is Br. 2,050.

- a. What is the estimated mean income of all middle managers (the population) ?
- b. What is the 95% confidence interval c(rounded to the nearest 10)
- c. What are the 95% confidence limits?

- d. Interpret the finding.

Solution

- a. Sample mean is 35 420 so this will approximate the population mean so $\mu = 35420$. It is estimated from the sample mean.
- b. The confidence interval is between 35170 and 35670 found by
- $$\bar{X} \pm 1.96 \frac{S}{\sqrt{n}} = 35420 \pm 1.96 \left(\frac{2050}{\sqrt{256}} \right) = 35168.87 \text{ and } 35671.13$$
- c. The end points of the confidence interval are called the confidence limits. In this case they are rounded to 35170 and 35670. 35170 is the lower limit and 35070 is the upper limit.
- d. Interpretation

If we select 100 samples of size 256 from the population of all middle managers and compute the sample means and confidence intervals, the mean annual income of the population would be found in about 95 out of the 100 confidence intervals. About 5 out of the 100 confidence intervals would not contain the population mean annual income.

Check Your Progress –2

A research firm conducted a survey to determine the mean amount smokers spend on cigarette during a week. A sample of 49 smokers revealed that the sample mean is Br. 20 with standard deviation of Br. 5. Construct 95% confidence interval for the mean amount spent.

Confidence interval for a population proportion

The confidence interval for a population proportion is estimated

$$\bar{p} \pm Z\sigma_p$$

Where σ_p is the standard error of the proportion and

$$\sigma_p = \sqrt{\frac{p(1-p)}{n}}$$

Therefore the confidence interval for population proportion is constructed by

$$\bar{p} \pm Z \sqrt{\frac{p(1-p)}{n}}$$

Example. Suppose 1600 of 2000 union members sampled said they plan to vote for the proposal to merge with a national union. Union, by laws state that at least 75% of all members must approve, for the merger to be enacted. Using the 0.95 degree of confidence, what is the interval estimate for the population proportion? Based on the confidence interval, what conclusion can be

drawn? $\bar{p} = \frac{1600}{2000} = 0.8$. The sample proportion is 80%

The interval is computed as follows.

$$\begin{aligned}\bar{p} \pm Z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} &= 0.80 \pm 1.96 \sqrt{\frac{0.80(1-0.8)}{2000}} = 0.80 \pm 1.96 \sqrt{0.00008} \\ &= 0.78247 \text{ and } 0.81753 \text{ rounded to } 0.782 \text{ and } 0.818.\end{aligned}$$

Based on the sample results when all union members vote, the proposal will probably pass because 0.75 lie below the interval between 0.782 and 0.818.

Check Your Progress –3

A sample of 200 people were assumed to identify their major source of news information; 110 stated that their major source was television news coverage. Construct a 90% confidence interval for the proportion of people in the population who consider television their major source of news information.

2.2.2 Finite Population Correction Factor

The population we have sampled so far has been very large, or assumed to be infinite.

If the sampled population is not infinite or not larger we need to make some adjustments in the standard error of the mean and the standard error of the proportion. This is done to reduce the error we committed in estimating a parameter.

A population that has a fixed upper bound is said to be finite. A finite population can be small or can be very large.

For a finite population, where the total number of objects is N , and the size of the sample is n the following adjustment is made to the standard errors of the mean and the proportion.

Standard error of the mean

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Standard error of the proportion

$$\sigma_{\bar{p}} = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \sqrt{\frac{N-n}{N-1}}$$

This adjustment is called **finite population correction factor**.

Why it is necessary to apply a factor and what is its effect?

Logically, if a sample is a substantial percentage of the population, then we would expect any estimate to be more precise than those for a smaller sample.

Suppose the population is 1000 and the sample is 100. Then this ratio is $\frac{1000-100}{1000-1}$ or $\frac{900}{999}$. Taking the square root gives the correction factor 0.9492. Multiplying the standard error reduces the error by about 5% or $(1-0.9492)=0.05$. This reduction of the size of the standard **error yields a smaller range** of values in estimating the population mean. If the sample size is 200 the correction factor is 0.8949. Meaning that the standard error has been reduced by more than 10%.

The usual rule is that If the ratio of the sample to the population, n/N , is less than 0.05, the finite population correction factor is ignored.

Example. There are 250 families in a small town A poll of 40 families revealed that the mean annual community contribution is 450 with a standard deviation of 75. Construct a 95% confidence interval for the mean annual contribution.

Solution:-

First note that the population is finite.

Second the sample constitute more than 5% of the population $n/N = 40/250 = 0.16$ Hence the finite population correction factor is applied.

$$\begin{aligned}\bar{x} \pm Z \frac{S}{\sqrt{n}} \left(\sqrt{\frac{N-n}{N-1}} \right) &= 450 \pm 1.96 \left(\frac{75}{\sqrt{40}} \right) \sqrt{\frac{250-40}{250-1}} \\ &= 450 \pm 23.24 \sqrt{0.8433} \\ &= 450 \pm 21.34 \\ &= 428.66 \text{ and } 471.34\end{aligned}$$

Confidence interval for small sample (Student's Distribution)

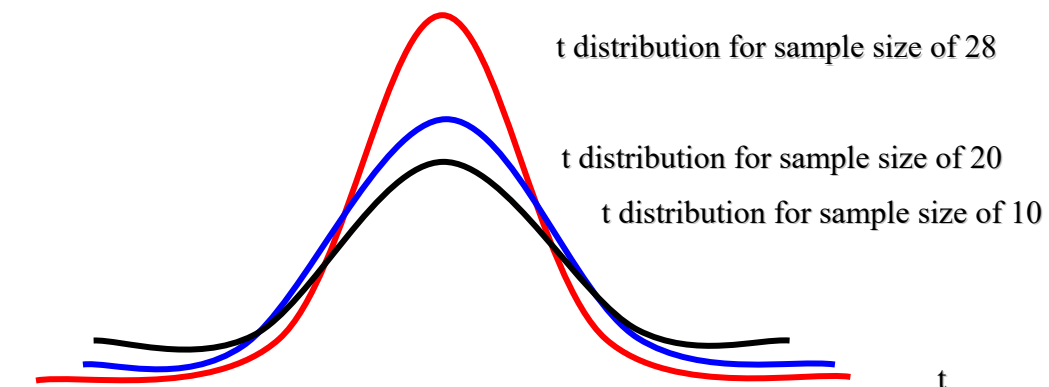
When the **population is large and normal and the standard deviation is known** the standard normal distribution is employed to construct the confidence interval for the mean and proportion. If the sample size is at least 30, the sample standard deviation can substitute the population standard deviation and the results are deemed satisfactory.

If the sample size is less than 30 and population standard deviation is unknown, the standard normal distribution, **Z**, is **not appropriate**. The student's t or the **t distribution** is used.

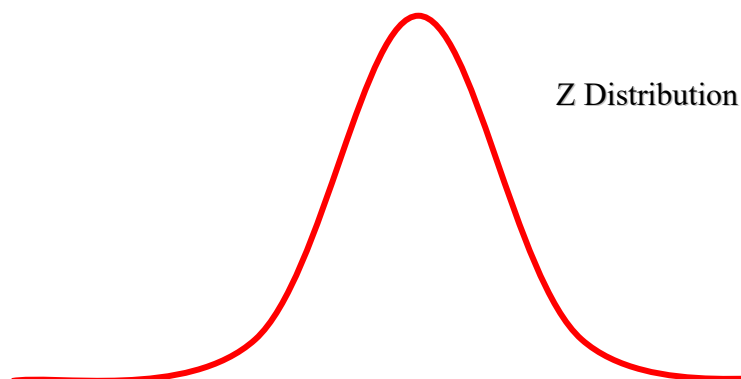
Characteristics of the Student's t Distribution

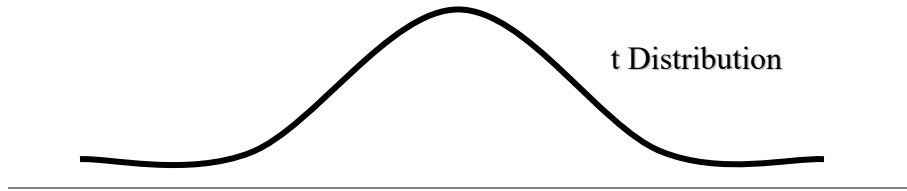
Assuming that the population of interest is normal or approximately normal, the following are the characteristics of the t distribution

1. It is a continuous distribution
2. It is bell-shaped and symmetrical
3. There is not one t distribution, but rather a 'family' of t distribution. All have the same mean of zero but their standard deviations differ according to the sample size, n. The t distribution differs for different sample size.
4. It is more spread out and flatter at the center than is the Z. However as the sample size increases the curve representing t distribution approaches the Z distribution.



As the sample size decreases the curve representing the t distribution will have wider tails and will be more flat at the center.





For a given confidence level, say 95%, the t value is greater than the Z value. This is so because there is **more variability in sample means computed from smaller samples**. Thus our confidence in the **resulting estimate is not strong**. t values are found referring to the appropriate **degrees of freedom in the t table**. Degree of freedom means the freedom to freely move data points or the freedom to freely assign values arbitrarily.

Degrees of freedom (df) = $n - 1$ where n is the sample size.

This implies that we can freely move or assign values for all data points except the last n^{th} value. If the mean of the distribution is specified there is a freedom to assign any value for all data points except the last point.

Example - the mean of five data points is 12. Then it follows that the sum of all the five points is $60 = (5 \times 12)$. Thus if five points are constrained to have a sum of 60 or a mean of 12, we have $5 - 1 = 4$ degrees of freedom.

If all the five data points are missing we are free to assign any value as long as their sum is 60 say 14, 12, 10, 9, 15. If 4 are missing we are free to assign any value since 60 minus the known value of a data point is known.

If two are unknown, 14, 16, 10, x_3 , x_4 since $14 + 16 + 10 + x_3 + x_4 = 60$

Then $x_3 + x_4 = 60 - 40 = 20$

$x_3 + x_4 = 20$. We can assign any value as long as their sum is 20. 10, 10 or 9, 11 or 15, 5 etc...

But if the four data points are known, (10, 14, 16, 12), the 5th data point will have a predetermined value i.e. $60 - 52 = 8$. Now we are not free to assign arbitrary value for this data point. Degrees of freedom can be obtained from the deviation based on the assumption that sum of the differences (d) between the mean and all values of the random variable (x) is zero. i.e., if we subtract the mean from all values of x the sum of the difference will be zero consider the

above five data points. Their mean is 12 and their sum 60. Thus $(x_1 - 12) + (x_2 + 12) + (x_3 - 12) + (x_4 - 12) + (x_5 - 12) = 0 = d_1 + d_2 + d_3 + d_4 + d_5 = 0$

Now we are free to assign any value for only four missing differences as long as this sum is zero. So we have still $n - 1$ degrees of freedom.

Computing t value

The t variable representing the student's t distribution is defined as

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} \quad \text{where: } \bar{x} \text{ is the sample mean of } n \text{ measurements, } \mu \text{ is the population mean and } s \text{ is the sample standard deviation}$$

Note that t is just like $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ except that we replace σ with s. unlike our methods of large samples, σ cannot be approximated by s when the sample size is less than 30 and we cannot use the normal distribution. The table for the **t distribution** is constructed for selected levels of confidence for degree of freedom up to 30. To use the table we need to know two numbers, the **tail area**, (1 minus confidence level selected), and the **degree of freedom**.

(1 – confidence level selected) is α , the Greek letter alpha. This is the error we committee in estimating.

The confidence interval for the sample mean is $\bar{x} \pm \alpha/2 (n-1) \frac{s}{\sqrt{n}}$

Example. A traffic department in town is planning to determine mean number of accidents at a high-risk intersection. Only a random sample of 10 days measurements were obtained.

Number of accidents per day were;

8, 7 10 15 11 6 8 5 13 12

Construct a 95% confidence interval for the mean number of accident per day.

a) Compute $\bar{x}$ and s

$$\bar{x} = \frac{95}{10} = 9.5 \text{ per day}$$

$$S_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\frac{94.5}{9}} = 3.24 \text{ per day}$$

The confidence level is 95% so

$$\alpha = 1 - 0.95 = 0.05$$

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$$

The degree of freedom, $df = n - 1 = 10 - 1 = 9$ from the t table $t_{0.025, df 9} = 2.76$

The confidence interval is

$$\begin{aligned} \bar{x} \pm t_{0.025, df(9)} \frac{s}{\sqrt{n}} \\ 9.5 \pm (2.26) \frac{3.24}{\sqrt{10}} \\ 9.5 \pm 2.3 \\ 7.2 \text{ to } 11.80 \end{aligned}$$

With 95% confidence the mean number of accident at this particular intersection is between 7.2 and 11.8.

Check Your Progress –4

A quality controller of a company plans to inspect the average diameter of small bolts made. A random sample of 6 bolts was selected. The sample is computed to be 2.0016mm and the sample standard deviation 0.0012mm. Construct the 99% confidence interval for all bolts made.

2.3 SELECTING A SAMPLE SIZE

Size of a sample must be determined scientifically. Care must be taken not to select a sample too large or too small. There are two misconceptions about how many to sample

- Sample consisting 5% (or similar constant percentage) is adequate for all problems.
5% can be too much for a particular population say 10 million or can be too small for another say 200.
- A sample, for example, must be selected from a heavily populated area.

To avoid such problems the sample size should be mathematically determined.

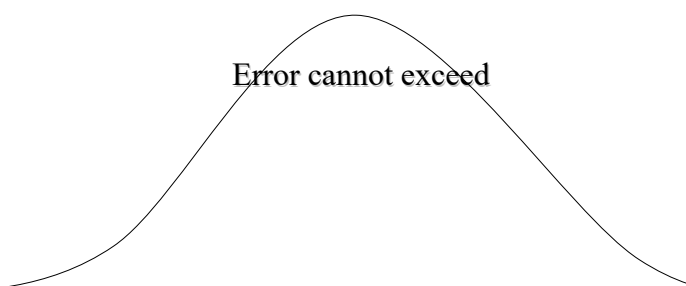
2.3.1 Sample Size for the Mean

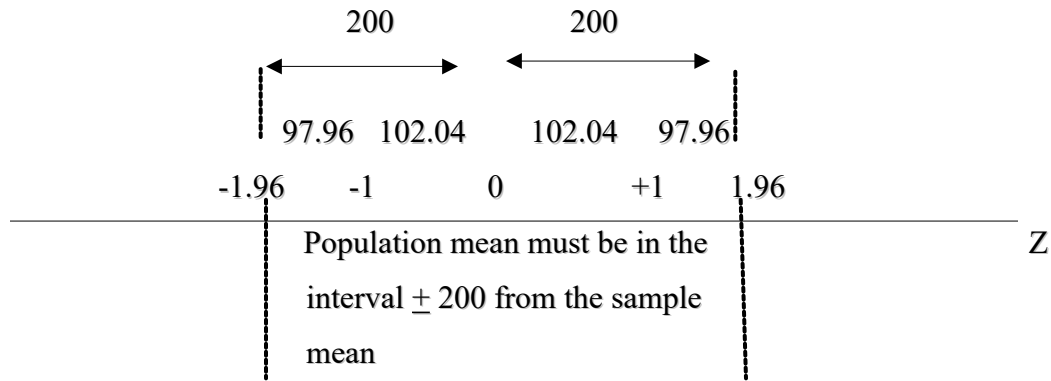
There are three factors that determine the size of the sample. None of which has any direct relationship to the size of the population.

- a. The degree of confidence selected.
- b. The maximum allowable error
- c. The variation in the population

- a. *The degree of confidence*, This is usually 95% or 99%. But it may be any level. It is specified by the statistician. The higher the degree of confidence, the larger the sample required. If we want to be sure the true mean will lie between an interval, we would have to survey the entire population. Example. Suppose the parameter to be estimated is the arithmetic mean, and the degree of confidence selected is 90%. Based on a sample, it was estimated that the population mean is in the interval between 850 and 1050. Logically, if the degree of confidence were increased to 95% or 99% the sample size would have to increase.
- b. *Maximum error allowed*. It is the maximum error that will be tolerable at a specified level of confidence. Suppose a statistician is interested to estimate the mean income of residents of an area. There are indications that the family incomes range from a probable low of 19000 to a high of about 39000. On the assumption that these are reasonable estimates, does it seem likely that the statistician would be satisfied with this statement resulting from a sample of area residents. “ The population mean is between 23,000 and 35,000” Probability not. Because confidence limits that wide indicate little or nothing about the population mean. Instead, the statistician stated “using the 0.95 confidence level, the total error is predicting the population mean should not exceed by 200”. The maximum allowable error is denoted ‘E’ = $E = |\bar{x} - \mu|$. This means based on a sample size n, if the estimate of population mean is computed to be 35,000, then we will assure that the population mean is in the interval between 34800 and 35200. Found by $35,000 + 200$ and $35,000 - 200$. For the 0.95 degree of confidence selected the maximum error of ± 200 in terms of Z is 1.96. To determine the value of one standard error of the mean $\sigma_{\bar{x}}$ simply divide the total error of 200 by 1.96 = 102.04

$$\sigma_{\bar{x}} = \frac{200}{1.96} = 102.04$$





The size of the sample is computed by solving for n in the formula

$$S_{\bar{x}} = \frac{S}{\sqrt{n}}, \text{ note that since we are using a sample standard deviation.}$$

i.e., $S_{\bar{x}}$ is substituted for $\sigma_{\bar{x}}$ and S for σ

$$\frac{\text{Total allowable error}}{Z \text{ standard deviation}} = \frac{\text{Sample standard deviation}}{\sqrt{\text{sample size}}}$$

Total allowable error let be represented by 'E' then it follows that,

$$\frac{E}{Z} = \frac{S}{\sqrt{n}} = \frac{200}{1.96} = \frac{S}{\sqrt{n}} = 102.04$$

Since there are two unknowns for one equation we cannot solve for both.

- c. *Variation in the population.* There are still two unknowns. To solve for the number to be sampled we need to estimate the variation in the population. The standard deviation is a measure of variation. Thus the standard deviation of the population must be estimated.

This can be done either:

- a. By taking a small pilot survey and using the standard deviation of the pilot sample as an estimate of the population standard deviation or
- b. By estimating the standard deviation based on knowledge of the population.

Suppose a pilot survey is conducted and sample standard deviation is computed to be 3000. The number to be sampled can now be estimated.

$$S_{\bar{x}} = \frac{S}{\sqrt{n}} = \frac{E}{Z}$$

$$\frac{200}{1.96} = \frac{3000}{\sqrt{n}} \quad n = 864.36$$

S_x is standard error of the mean, the error we commit in estimating . From the above computation we can learn that as the variation in the population increase the sample size will increase.

A more convenient computational formula for determining n is.

$$n = \left(\frac{Z \cdot S}{E} \right)^2$$

where E = allowable error

Z = Z value for the degree of confidence selected

S = Sample deviation

For this example $n = \left(\frac{1.96 \times 3000}{200} \right)^2 = 864.36$

Example 1. A marketing research firm wants to conduct a survey to estimate the average amount spent on entertainment by each person visiting a popular pub. The people who plan the survey would like to be able to determine the average amount spent by all people visiting the pub to within br. 120, with 95% confidence. From past operations of the pub, an estimate of the population standard deviation is $\sigma = \text{br. } 400$ what is the minimum required sample sizes?

$$Z = 1.96$$

$$E = 120$$

$$\sigma = 400$$

Required, n?

$$n = \left(\frac{1.96 \times 400}{120} \right)^2 = 42.68 \approx 43$$

Check Your Progress –5

A processor of carrots cuts the green top of each carrot washes the carrots, and inserts six to a package. Twenty packages are inserted in a box for shipment. To test the Wight of the boxes, a few were checked. The mean weight was 10kg and the standard deviation 0.25kg. How many

boxes must the processor sample to be 95% confident that the sample mean does not differ from the population mean by more than 0.1 kg?

2.3.2 Sample size for proportion

The procedure used to determine the sample size for the mean is applicable to determine when proportions are involved.

Three things must be specified.

- Decide on the level of confidence
- Indicate how precise the estimate of the population proportion must be
- Approximate the population proportion, P , either from past experience or from a small pilot survey $\bar{p}$

The formula for determining the sample size n for a proportion

$$n = \frac{\bar{p}(1 - \bar{p})}{E^2} Z^2$$

where: $\bar{p}$ - estimated proportion

Z = Z value for the selected confidence level

E = the maximum tolerable error

Example 1. A member of parliament wants to determine her popularity in her a region. She indicates that the proportion of voters who will vote for her must be estimated with in ± 2 percent of the population proportion. Further, the 95% degree of confidence is to be used. In past elections she received 40% of the popular vote in that area. She doubts whether it has changed much. How many registered voters should be sampled?

$$Z = 1.96$$

$$\bar{p} = 0.40$$

$$E = 0.02$$

$$\begin{aligned} n &= \frac{\bar{p}(1 - \bar{p})}{E^2} Z^2 \\ &= 0.40(1 - 0.4) \left(\frac{1.96}{0.02} \right)^2 = 2,304.96 \approx 2305 \end{aligned}$$

This sample size might be too large, or too small or exactly correct depending on the accuracy of $\bar{p}$.

Note: if there is no logical estimate of $\bar{p}$, the sample size can be estimated by letting $\bar{p}=0.5$

Example 2. Suppose the president wants an estimate of the proportion of the population that support this current policy on unemployment. The president wants the estimate to be within 0.04 of the true proportion. Assume a 95% level of confidence and the proportion supporting current policy to be 0.60.

- a) How large a sample is required
- b) How large would the sample have to be if the estimate were not available?

Solution:

$$\begin{array}{ll} \text{a) } E = 0.04 & \\ Z = 1.96 & n = 0.6(1 - 0.6) \left(\frac{1.96}{0.04} \right)^2 \\ \bar{p} = 0.60 & = \underline{577} \end{array}$$

$$\begin{array}{ll} \text{b) } E = 0.04 & \\ Z = 1.96 & \\ \bar{p} = 0.50 & \text{(since there is no estimate)} \end{array}$$

$$n = 0.5 (1 - 0.5) \left(\frac{1.96}{0.04} \right)^2 = 600$$

Check Your Progress –6

The marketing department of a company wishes to study the loyalty pattern of consumers. Loyalty patterns range from extremely loyal to brand snitcher. If the department wishes to estimate the proportion of consumers who are extremely loyal to this brand, what sample size would be necessary to estimate this proportion with 0.05 with 95% confidence?